Phone Access and Login

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Users could open our app on their phone over mobile data, from anywhere, and log in with a username and password that the Admin creates. This is possible, but campus IT must agree and the login must be strong. We decide it together at the next meeting.

The app on phone and PC

Our software has two halves, and only one of them goes on the phone.

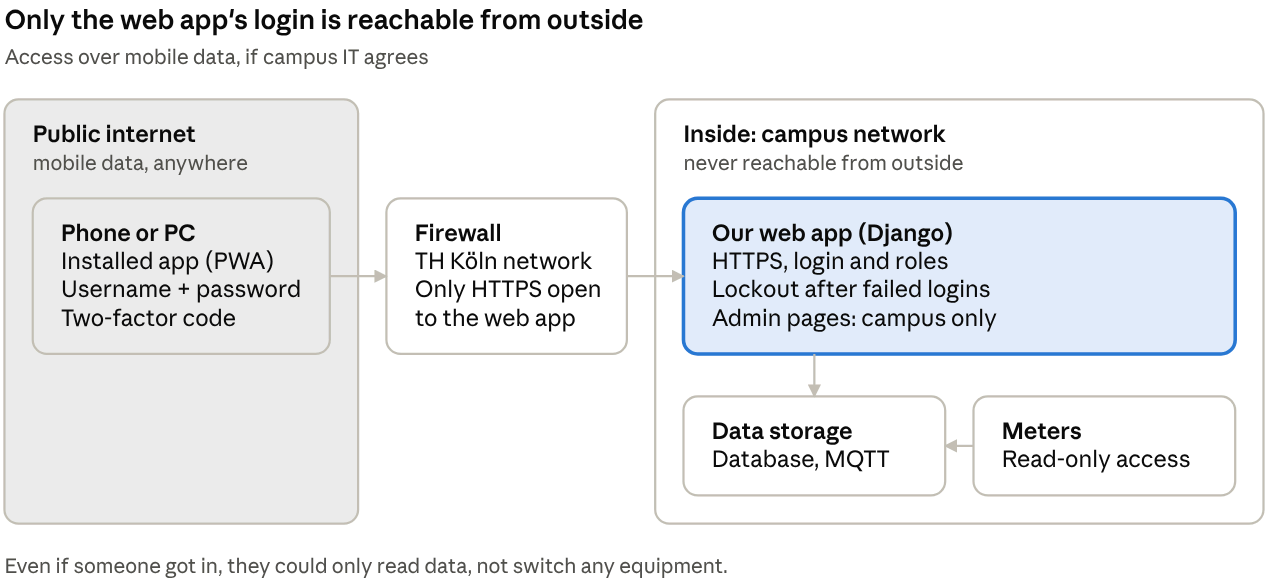
Half	What it does	Where it runs
Server	Collects and stores the meter data, runs the health check, reports and machine learning	Our PCs during development; a campus server later. Never on the phone.
App	Dashboards, alerts and buttons that users see	Phone and PC, as an installable web app (PWA)

The phone works like a window onto the server, the same way a banking or e-mail app does. Users open a link, tap **Install** or **Add to Home Screen**, and get an app icon that opens full-screen and can receive push alerts. The phone needs no Python, no Docker and no setup, and we write one set of pages that adapts to both screen sizes.

Login and network: two different things

- **Username and password decide who may use the app.** This is the user management on our poster: the Admin creates an account, gives it a role (Admin, Operator or Viewer), and deletes it when someone leaves. It works the same on Wi-Fi or mobile data.
- **The network decides whether the phone can reach the server at all.** A password only helps once the phone can find the server. On mobile data the phone comes from the public internet, so the server needs a public address, for example something like `https://gm-power.th-koeln.de`, and campus IT has to allow that.

How access from anywhere would work



Access from anywhere · phone, firewall, web app and the campus network

The phone reaches only our web app, through the university firewall. The meters, the database and the MQTT broker stay inside the campus network and are never reachable from the internet, the same way the university's webmail or learning platform works.

Protections a public login needs

If the login page is reachable from the internet, anyone can try to log in, so the login must be strong.

Protection	What it does	How we would do it
HTTPS only	Passwords travel encrypted	A certificate from the university, or Let's Encrypt
Account lockout	Blocks password guessing	django-axes, already installed
Strong password rules	No short or common passwords	Built into Django
Two-factor login	A stolen password alone is not enough: users also enter a code from an authenticator app	Strongly recommended for a public app

Protection	What it does	How we would do it
Session timeout	Logs out inactive users	Django setting
Admin pages hidden	User management only reachable inside the campus network	Web server rule
Read-only system	The app can only show data; it cannot switch any equipment on campus	Our design

The read-only design is our strongest argument with campus IT: even if someone broke in, they could only read data.

Questions for the meeting

For the team

#	Question
1	Should users reach the app from anywhere on mobile data, or only on campus?
2	If from anywhere, do we make two-factor login compulsory?

For campus IT

#	Question
1	Can our web app be reachable from the internet, with login, like other university services?
2	Can we get a domain name (such as `gm-power.th-koeln.de`) and an HTTPS certificate?
3	Can the firewall be opened for the app only (port 443), with the meters and database staying internal?

#	Question
4	Do they prefer university accounts (single sign-on) instead of our own usernames?
5	Can they give us a small server or virtual machine on the campus network?

None of this blocks development: until these are answered, everything runs on our own PCs.